

Graphs

Slides based on those of Kenneth H. Rosen

Slides by Sylvia Sorkin, Community College of Baltimore County - Essex Campus

Graphs

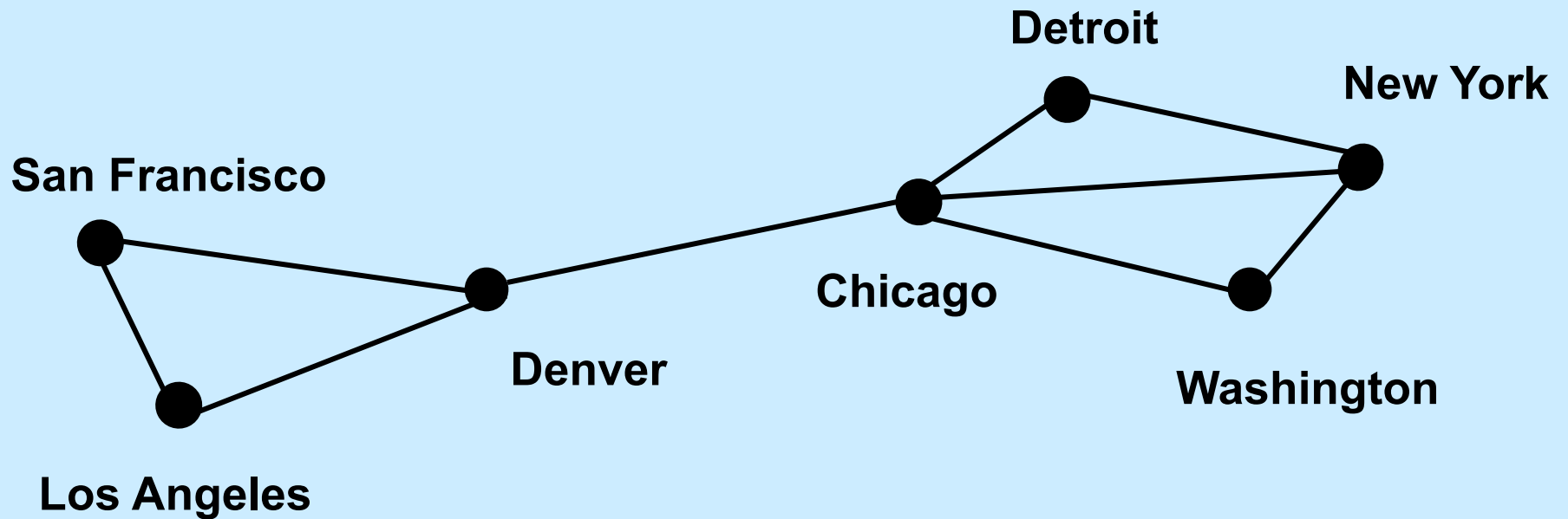
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Simple Graph

Definition 1. A **simple graph** $G = (V, E)$ consists of V , a nonempty set of **vertices**, and E , a set of unordered pairs of distinct elements of V called **edges**.

A simple graph



How many vertices? How many edges?

A simple graph

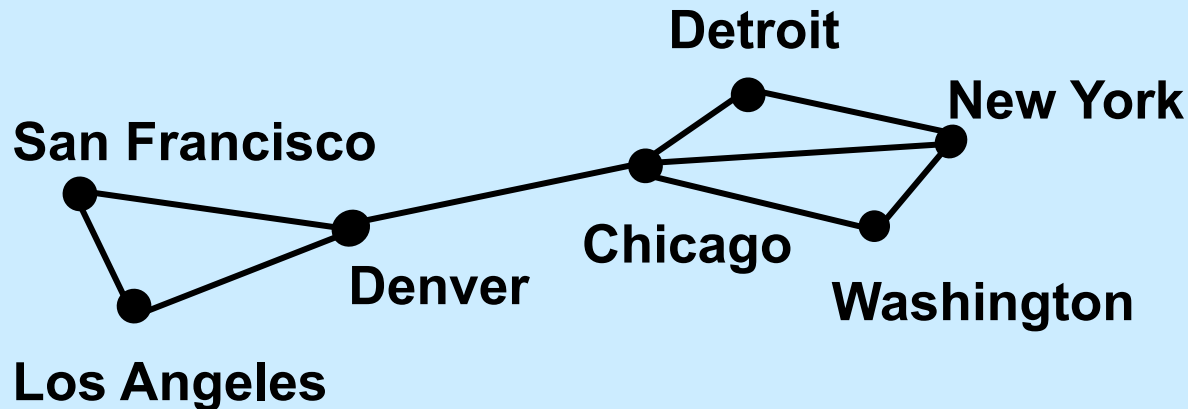
SET OF VERTICES

$V = \{ \text{Chicago, Denver, Detroit, Los Angeles, New York, San Francisco, Washington} \}$

SET OF EDGES

$E = \{ \{ \text{San Francisco, Los Angeles} \}, \{ \text{San Francisco, Denver} \}, \{ \text{Los Angeles, Denver} \}, \{ \text{Denver, Chicago} \}, \{ \text{Chicago, Detroit} \}, \{ \text{Detroit, New York} \}, \{ \text{New York, Washington} \}, \{ \text{Chicago, Washington} \}, \{ \text{Chicago, New York} \} \}$

A simple graph



The network is made up of computers and telephone lines between computers. There is at most 1 telephone line between 2 computers in the network. Each line operates in both directions. No computer has a telephone line to itself.

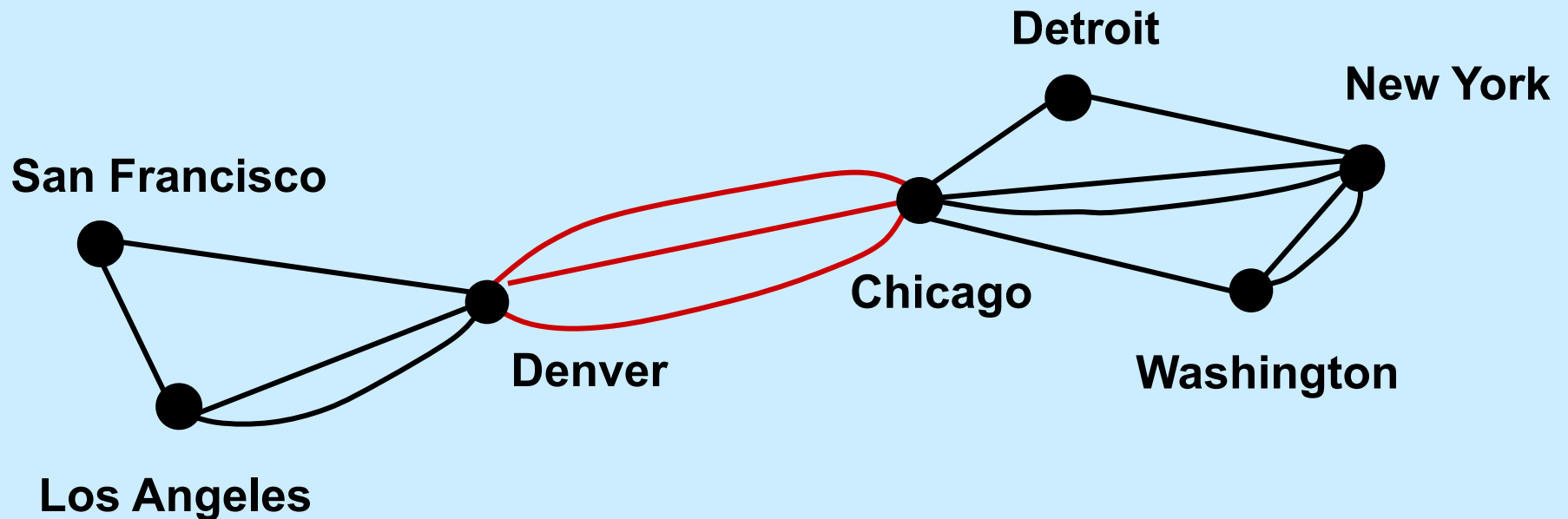
These are undirected edges, each of which connects two distinct vertices, and no two edges connect the same pair of vertices.

A Non-Simple Graph

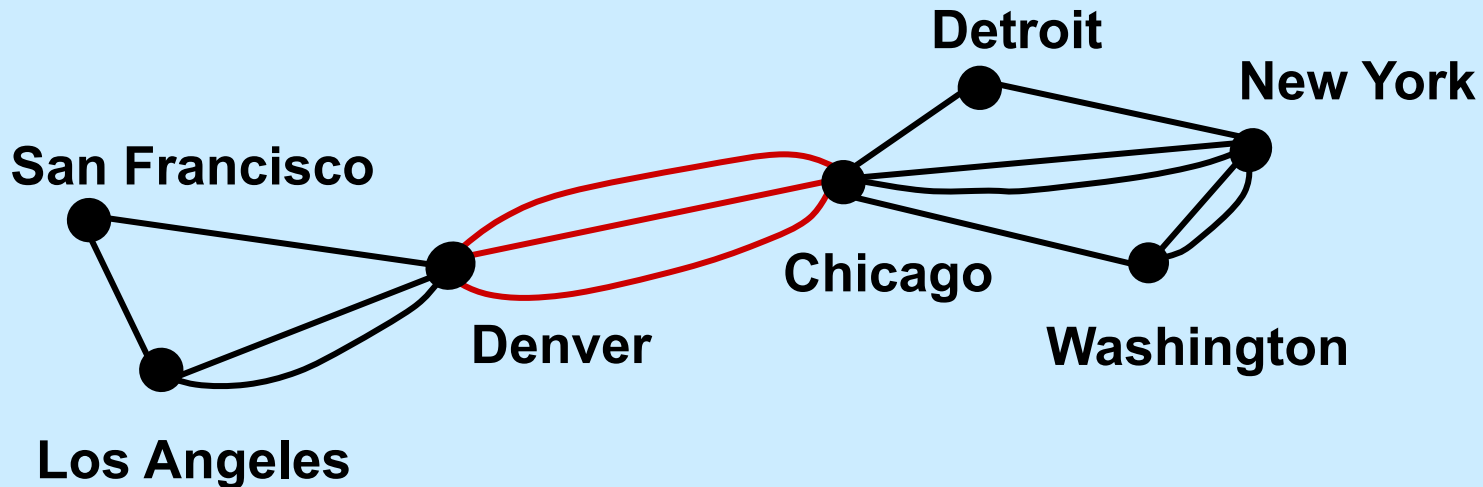
Definition 2. In a **multigraph** $G = (V, E)$ two or more edges may connect the same pair of vertices.

A Multigraph

**THERE CAN BE MULTIPLE TELEPHONE LINES
BETWEEN TWO COMPUTERS IN THE NETWORK.**



Multiple Edges



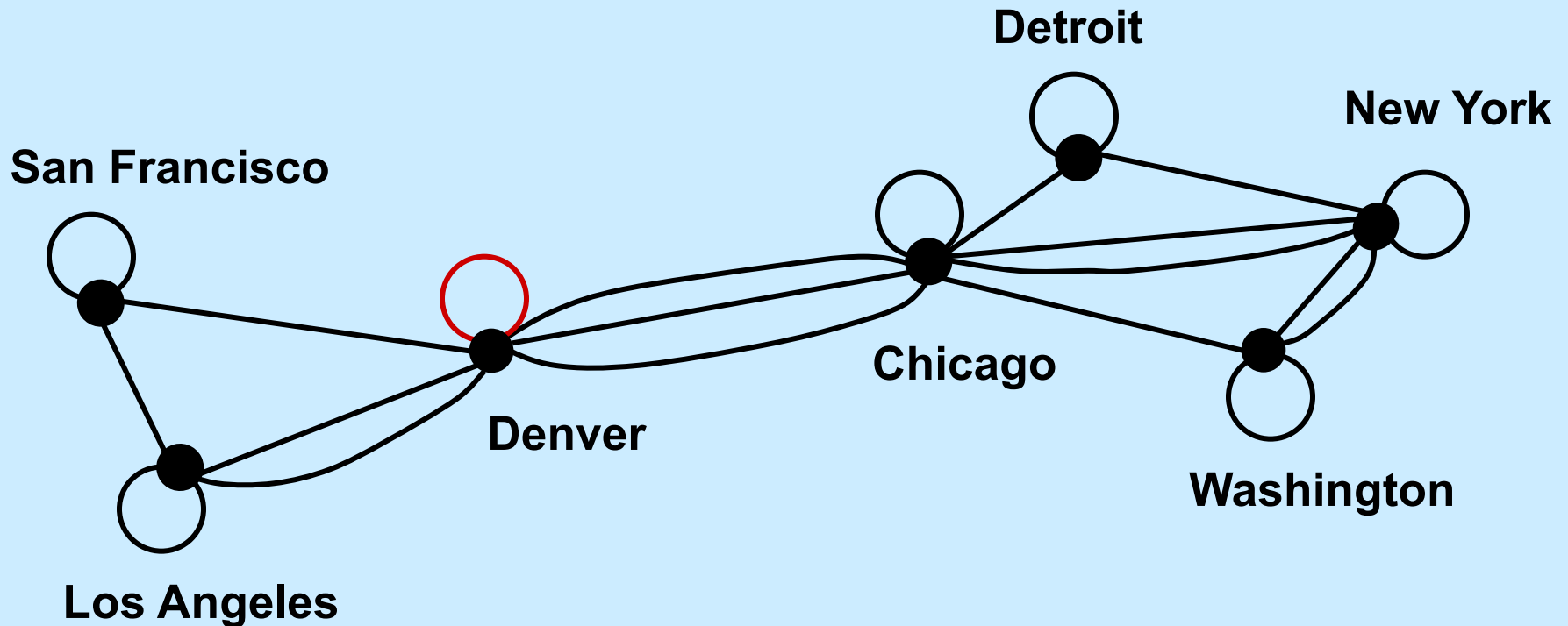
Two edges are called *multiple or parallel edges* if they connect the same two distinct vertices.

Another Non-Simple Graph

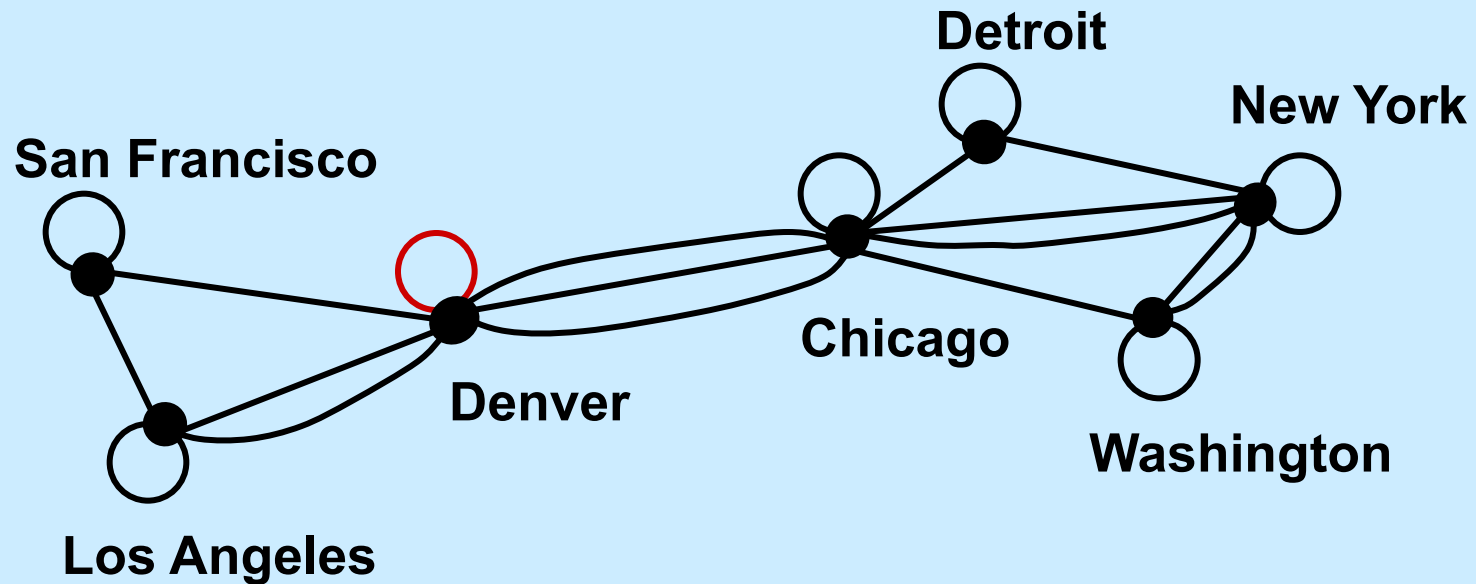
Definition 3. In a **pseudograph** $G = (V, E)$ two or more edges may connect the same pair of vertices, and in addition, an edge may connect a vertex to itself.

A Pseudograph

**THERE CAN BE TELEPHONE LINES IN THE NETWORK
FROM A COMPUTER TO ITSELF (for diagnostic use).**

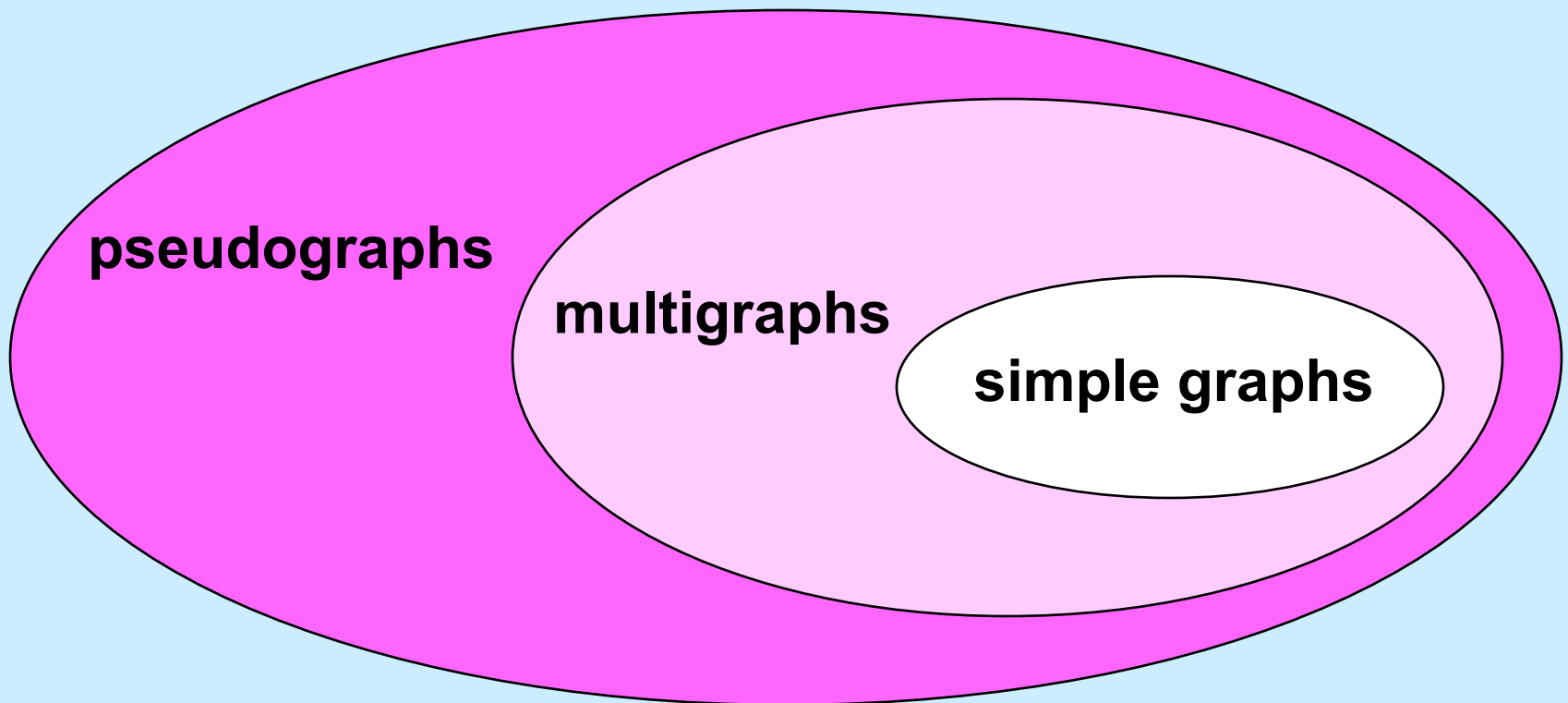


Loops



An edge is called a *loop* if it connects a vertex to itself.

Undirected Graphs



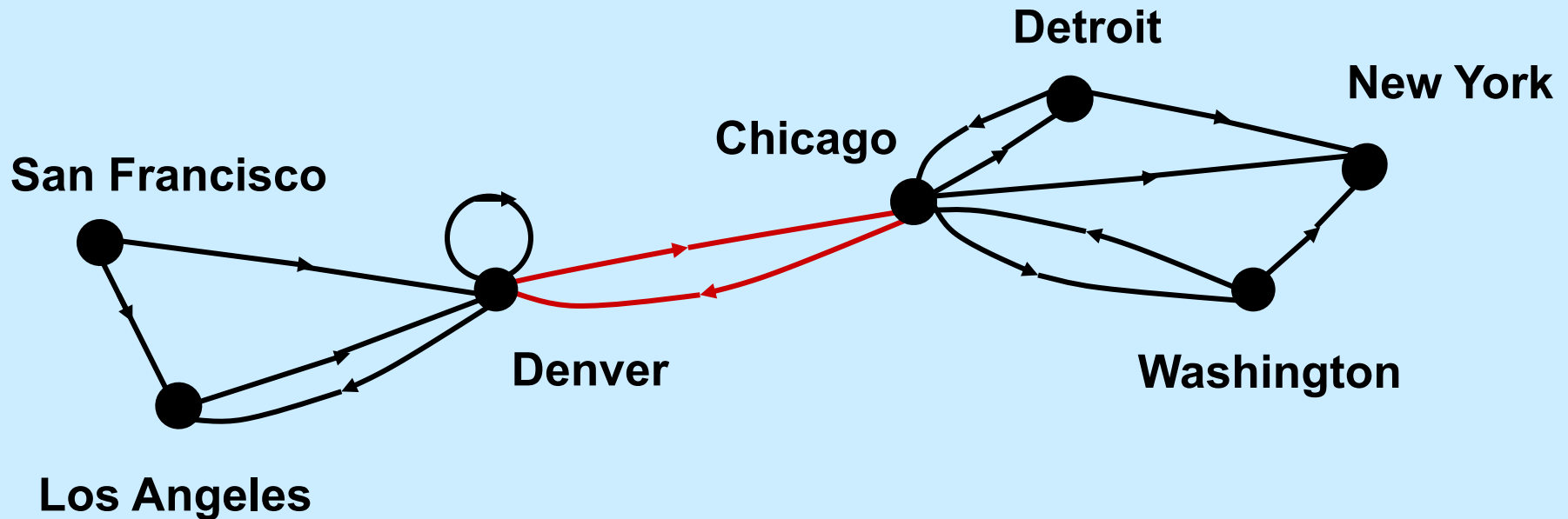
A Directed Graph

Definition 4. In a **directed graph** $G = (V, E)$ the edges are ordered pairs of (not necessarily distinct) vertices.

A Directed Graph

**SOME TELEPHONE LINES IN THE NETWORK
MAY OPERATE IN ONLY ONE DIRECTION .**

**Those that operate in two directions are represented
by pairs of edges in opposite directions.**

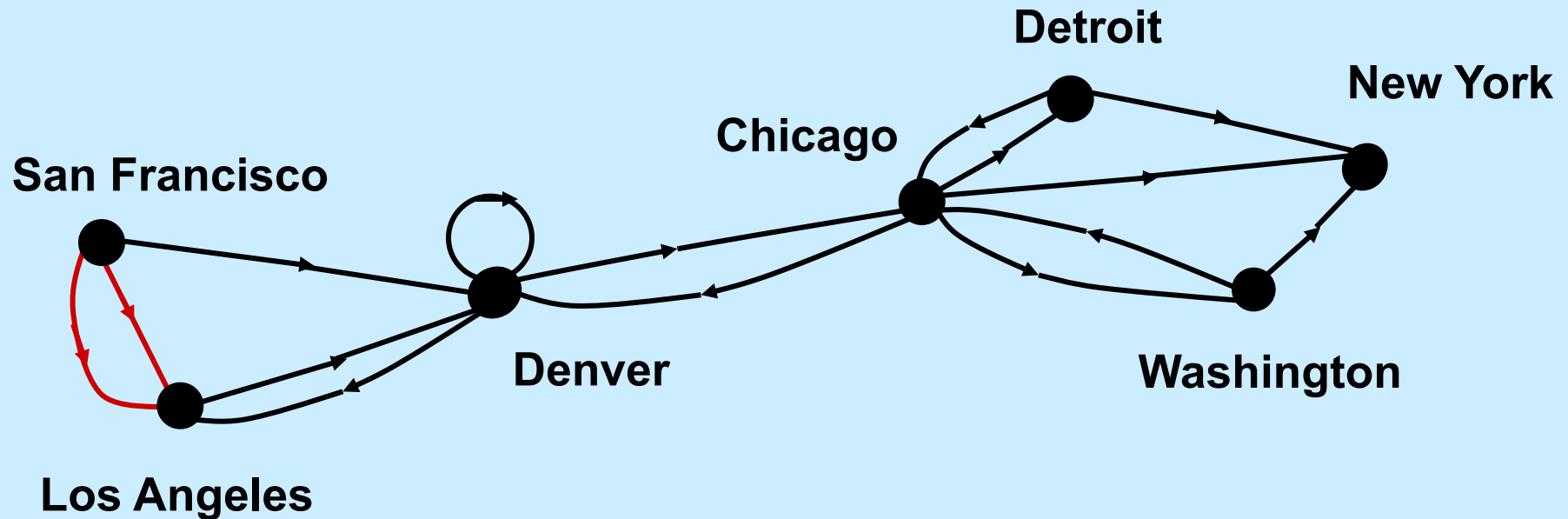


A Directed Multigraph

Definition 5. In a **directed multigraph** $G = (V, E)$ the edges are ordered pairs of (not necessarily distinct) vertices, and in addition there may be multiple edges.

A Directed Multigraph

**THERE MAY BE SEVERAL ONE-WAY LINES
IN THE SAME DIRECTION FROM ONE COMPUTER
TO ANOTHER IN THE NETWORK.**



Types of Graphs

TYPE	EDGES	MULTIPLE EDGES ALLOWED?	LOOPS ALLOWED?
Simple graph	Undirected	NO	NO
Multigraph	Undirected	YES	NO
Pseudograph	Undirected	YES	YES
Directed graph	Directed	NO	YES
Directed multigraph	Directed	YES	YES

Adjacent Vertices (Neighbors)

Definition 1. Two vertices, u and v in an undirected graph G are called **adjacent** (or **neighbors**) in G , if $\{u, v\}$ is an edge of G .

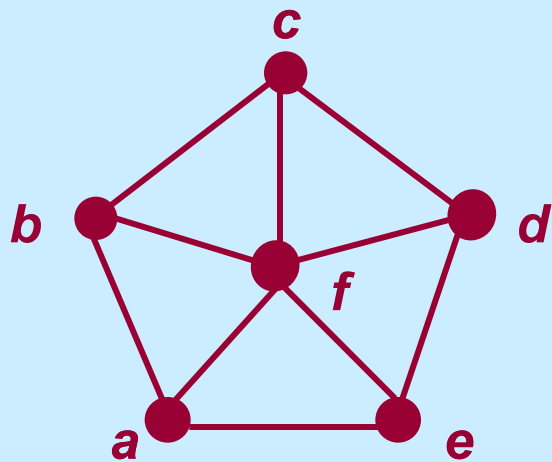
An edge e connecting u and v is called **incident with vertices u and v** , or is said to connect u and v . The vertices u and v are called **endpoints** of edge $\{u, v\}$.

Adjacency Matrix

A simple graph $G = (V, E)$ with n vertices can be represented by its adjacency matrix, A , where entry a_{ij} in row i and column j is

$$a_{ij} = \begin{cases} 1 & \text{if } \{v_i, v_j\} \text{ is an edge in } G, \\ 0 & \text{otherwise.} \end{cases}$$

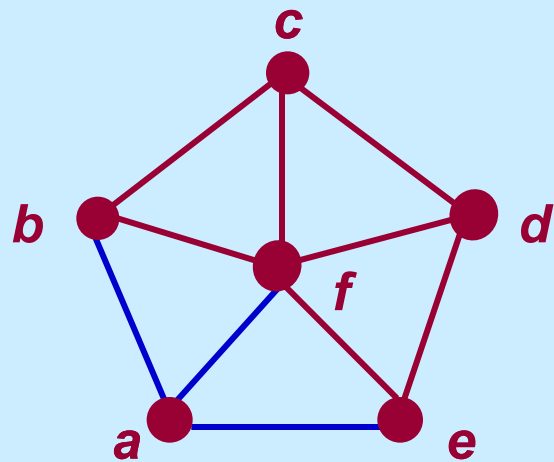
Finding the adjacency matrix



W_5

This graph has 6 vertices *a*, *b*, *c*, *d*, *e*, *f*. We can arrange them in that order.

Finding the adjacency matrix



W_5

FROM

TO

a b c d e f

a

0 1 0 0 1 1

b

c

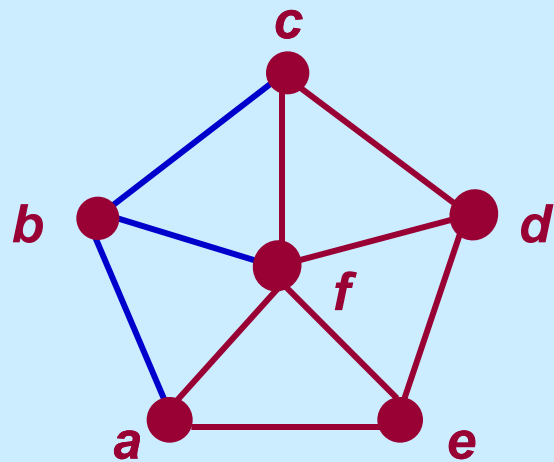
d

e

f

There are edges from a to b, from a to e, and from a to f

Finding the adjacency matrix



W_5

FROM

TO

a b c d e f

a

0 1 0 0 1 1

b

1 0 1 0 0 1

c

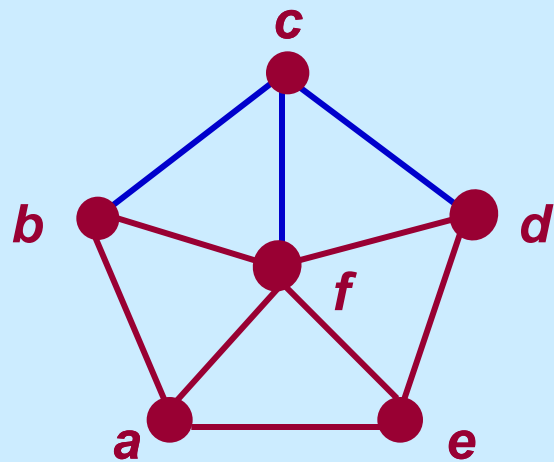
d

e

f

There are edges from b to a, from b to c, and from b to f

Finding the adjacency matrix



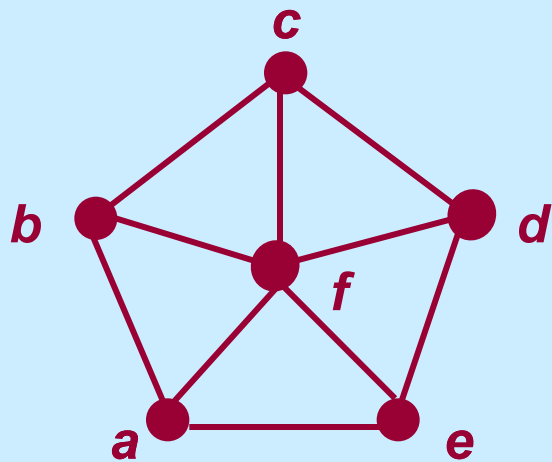
FROM

TO

	a	b	c	d	e	f
a	0	1	0	0	1	1
b	1	0	1	0	0	1
c	0	1	0	1	0	1
d						
e						
f						

There are edges from c to b, from c to d, and from c to f

Finding the adjacency matrix



W_5

FROM

TO

a *b* *c* *d* *e* *f*

a

0 1 0 0 1 1

b

1 0 1 0 0 1

c

0 1 0 1 0 1

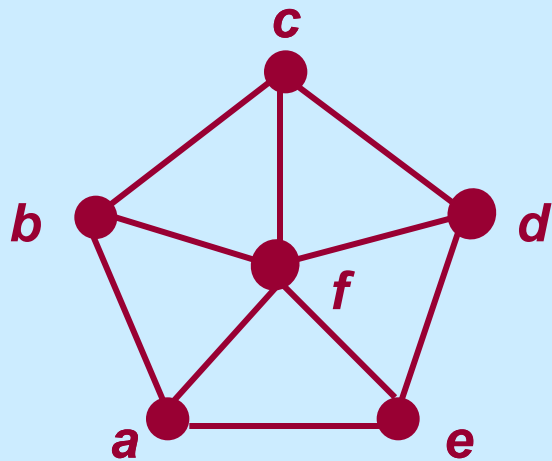
d

e

f

COMPLETE THE ADJACENCY MATRIX . . .

Finding the adjacency matrix



W_5

FROM

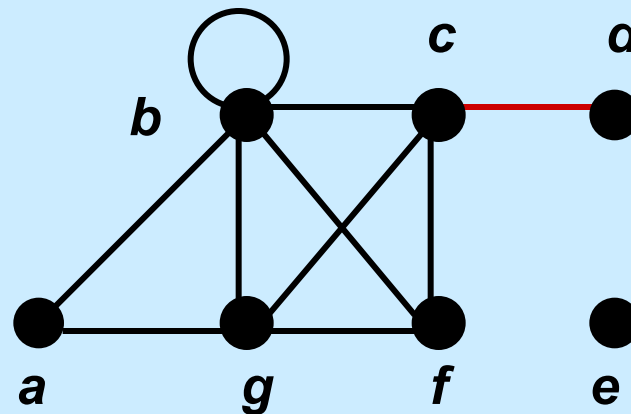
TO

	a	b	c	d	e	f
a	0	1	0	0	1	1
b	1	0	1	0	0	1
c	0	1	0	1	0	1
d	0	0	1	0	1	1
e	1	0	0	1	0	1
f	1	1	1	1	1	0

Notice that this matrix is symmetric. That is $a_{ij} = a_{ji}$ Why?

Degree of a vertex

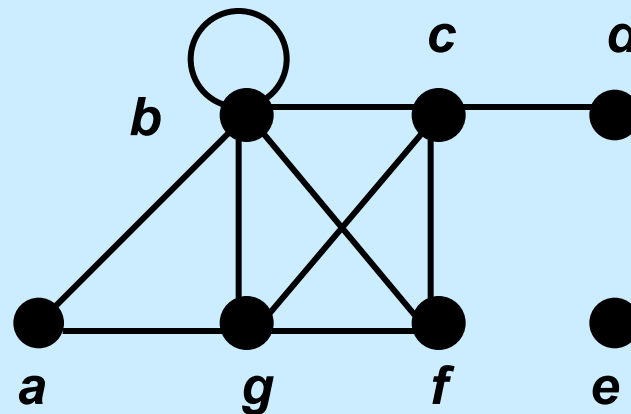
Definition 1. The **degree of a vertex** in an undirected graph is the number of edges incident with it, except that a loop at a vertex contributes twice to the degree of that vertex.



$$\deg(d) = 1$$

Degree of a vertex

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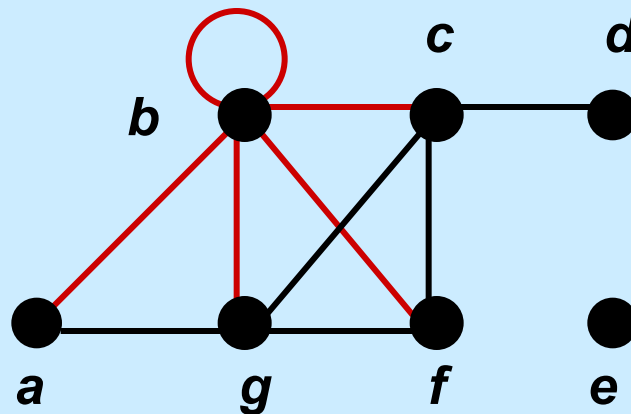


$$\deg(e) = 0$$

Degree of a vertex

Definition 1. The **degree of a vertex** in an undirected graph is the number of edges incident with it, except that a loop at a vertex contributes twice to the degree of that vertex.

$$\deg(b) = 6$$



Degree of a vertex

Find the degree of all the other vertices.

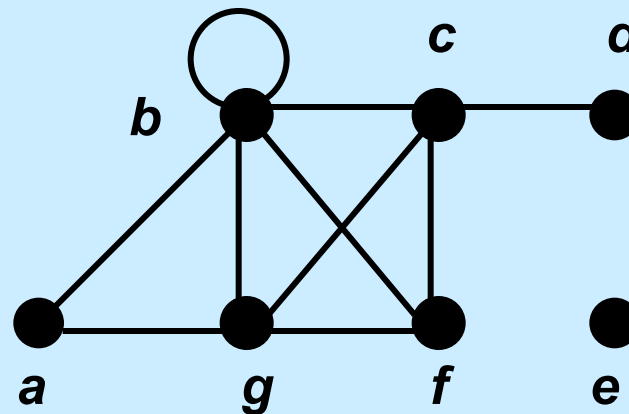
$\deg(a)$

$\deg(c)$

$\deg(f)$

$\deg(g)$

$\deg(b) = 6$



$\deg(d) = 1$

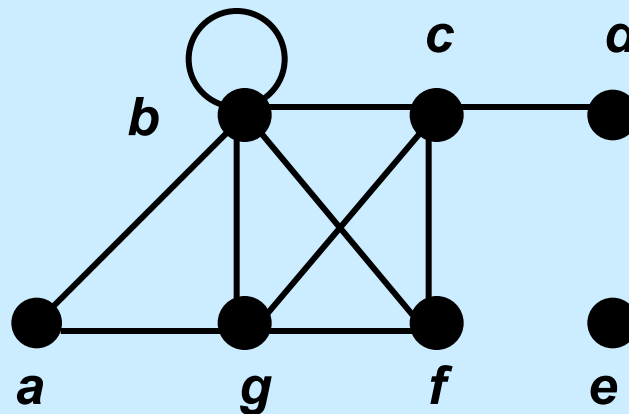
$\deg(e) = 0$

Degree of a vertex

Find the degree of all the other vertices.

$\deg(a) = 2$ $\deg(c) = 4$ $\deg(f) = 3$ $\deg(g) = 4$

$\deg(b) = 6$



$\deg(d) = 1$

$\deg(e) = 0$

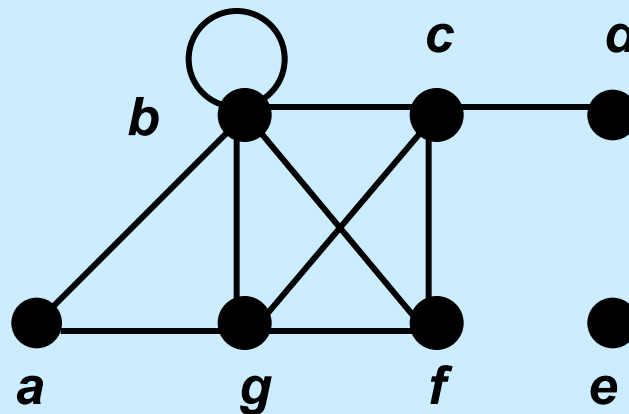
Degree of a vertex

Find the degree of all the other vertices.

$$\deg(a) = 2 \quad \deg(c) = 4 \quad \deg(f) = 3 \quad \deg(g) = 4$$

$$\text{TOTAL of degrees} = 2 + 4 + 3 + 4 + 6 + 1 + 0 = 20$$

$$\deg(b) = 6$$



$$\deg(d) = 1$$

$$\deg(e) = 0$$

Degree of a vertex

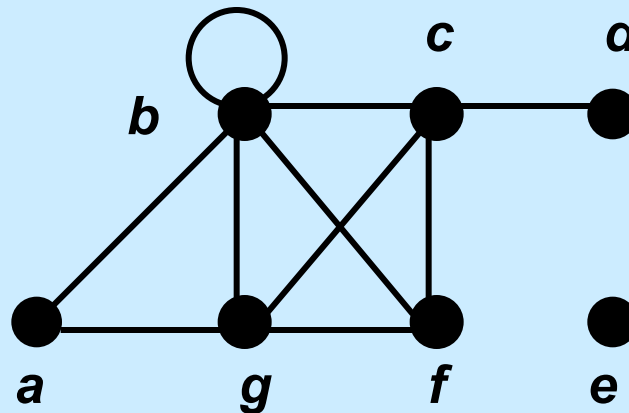
Find the degree of all the other vertices.

$$\deg(a) = 2 \quad \deg(c) = 4 \quad \deg(f) = 3 \quad \deg(g) = 4$$

$$\text{TOTAL of degrees} = 2 + 4 + 3 + 4 + 6 + 1 + 0 = 20$$

$$\text{TOTAL NUMBER OF EDGES} = 10$$

$$\deg(b) = 6$$



$$\deg(d) = 1$$

$$\deg(e) = 0$$

Handshaking Theorem

Theorem 1. Let $G = (V, E)$ be an undirected graph G with e edges. Then

$$\sum_{v \in V} \deg(v) = 2e$$

“The sum of the degrees over all the vertices equals twice the number of edges.”

NOTE: This applies even if multiple edges and loops are present.